

SARMAD ALI

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PROFESSIONAL SUMMARY

Data Science graduate from FAST NUCES Islamabad with hands-on experience designing, training, and deploying machine learning models, LLM-powered systems, and end-to-end AI pipelines. Proficient in Python, PyTorch, TensorFlow, and Scikit-learn across the full ML lifecycle, spanning data preprocessing, model training, evaluation, optimisation, and production deployment. Focused on solving real-world business problems through applied AI research.

EXPERIENCE

Full Stack Media Engineer, Speeqr

2026 | Remote | Present (3 Months)

- Debug audio sync issues across multi-room conferencing using NTP/GCC-PHAT alignment; fix a 16kHz/48kHz sample rate mismatch causing buffer overflow.
- Implement an adaptive jitter buffer and reduce end-to-end audio latency from 3–4 seconds to under 1 second through targeted tuning and AudioContext optimisation.
- Research WebSocket, WebRTC, and WebTransport/QUIC tradeoffs to inform architecture decisions on a .NET 9/C# + WebRTC stack; access and test deployments on AWS EC2 instances via RDP for remote debugging.

AI Automation Engineer, Escaplix

2025 | Islamabad | 3 Months

- Designed LLM-based automation workflows to eliminate repetitive operational tasks; integrated language-model APIs with internal tooling for intelligent, context-aware processing.

AI & ML Freelancer, Self-Employed

2022 – Present | Remote

- Delivered classification and regression models via REST APIs; engineered data pipelines, automated reporting, and insight dashboards for multiple client engagements.

Customer Service Representative, ARXA Communication & KAACHO

2022 – 2023 | Islamabad

- Managed client-facing operations and CRM workflows over two roles, strengthening professional communication and problem-resolution skills.

FINAL YEAR PROJECT

SkinVisionAI, AI-Powered Skin Disease Diagnosis System

2025 – 2026

FAST NUCES | ConvNeXt-V2 · PyTorch · FastAPI · PostgreSQL · JWT · Google Gemini · OpenRouter · Groq

- Trained and served a ConvNeXt-V2 model classifying 11 skin diseases at sub-second GPU inference; built an ROI detection stage using YCrCb colour space, Gaussian filtering, and morphological operations.
- Integrated RAG-style report generation grounded in a structured medical knowledge base, with automatic fallback over three LLM providers; validated end-to-end through 23 unit tests.

KEY PROJECTS

WGAN-GP Pneumonia Detection, Generative AI & Ensemble Learning

2026

WGAN-GP · TensorFlow · VGG16 · ResNet50 · Transfer Learning · IEEE Conference Paper

- Novel two-stage framework: WGAN-GP generates synthetic chest X-rays to resolve class imbalance; a VGG16 + ResNet50 transfer learning ensemble performs binary pneumonia classification, published as a full IEEE Conference-format research paper.

Optimised ML Pipeline, GPU-Accelerated Binary Classification

2024

PyTorch · CUDA · Python Multiprocessing · Scikit-learn · Benchmarking

- Modular PyTorch pipeline with GPU acceleration and parallel multiprocessing cutting runtime by over 70%; benchmarked CPU/GPU and serial/parallel configurations, reporting accuracy alongside wall-clock cost.

IoT Intrusion Detection, ML Benchmark Pipeline

2024

Scikit-learn · Random Forest · Linear SVC · KNN · Decision Tree · CICIoT2023

- Reproducible end-to-end evaluation comparing four classifiers on CICIoT2023 spanning 33 cyberattack types, with rigorous per-class precision, recall, and F1 reporting.

Electric Load Forecasting, Predictive Modelling Pipeline

2024

XGBoost · LSTM · Random Forest · K-Means · DBSCAN · Isolation Forest

- Forecasted next-day electricity demand for 10 US cities; applied anomaly detection, unsupervised clustering, and stacked ensemble methods to outperform naive baselines on MAE and RMSE.

Signature Verification · Image Moderation · Music Recommendation

2024

TensorFlow · MLP · FastAPI · Docker · MongoDB Atlas · Spotify API · Content-Based Filtering

- [Signature Verification] Binary genuine/forged classification via TensorFlow MLP with Otsu binarisation and 9-dimensional geometric feature extraction.
- [Image Moderation] AI safety scoring service for image URLs via REST endpoint; containerised with Docker Compose and documented in Swagger.
- [Music Recommendation] Content-based filtering engine on Spotify data with iterative user feedback loop to continuously refine model outputs.

EDUCATION

BS Data Science, FAST NUCES Islamabad

2022 – 2026

Coursework: Machine Learning, Deep Learning, NLP, Data Mining, Statistics, Databases, Software Engineering, Data Structures & Algorithms

TECHNICAL SKILLS

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face, JAX (familiar)

AI & Data: LLMs, RAG Pipelines, Prompt Engineering, GANs, Transfer Learning, NLP, CNNs, Feature Engineering, EDA

Languages & Tools: Python (primary), C++, SQL, JavaScript · Docker, FastAPI, Flask, Git, Jupyter, Postman, Linux, Pandas, NumPy

Databases & Cloud: PostgreSQL, MongoDB Atlas, MySQL, Firebase · Google Gemini API, OpenRouter, Groq

SOFT SKILLS

Problem Solving · Research Mindset · Attention to Detail · Team Collaboration · Agile Mindset · Communication · Leadership